

Automating the Order-to-Cash Process of a B2B Food Wholesaler: A Combined RPA and Agentic AI Proposal

1. Process Identification

1.1 Domain and organisational context

The process selected for this assessment is the *Order-to-Cash* process of WholesaleCo Foods Ltd., a fictitious but representative small-to-medium enterprise (SME) distributing dry goods, oils and chilled products to restaurants and independent retailers in the Dublin area. The company, its customers and all data used in this report are invented for the purposes of this assessment; any resemblance to an existing business is coincidental. Wholesale distributors of this kind operate as commercial intermediaries between the production of goods and retail trade, serving industrial enterprises, other wholesalers and retailers [1]. Their competitive position therefore rests less on the products themselves — which are largely undifferentiated commodities — than on the speed, accuracy and reliability of the order-handling process that surrounds them.

1.2 The process and why it matters

Order-to-Cash spans every activity from the receipt of a customer order to the collection of payment for it: order capture, stock verification, pricing and discounting, order confirmation, picking, delivery, invoicing and receivables follow-up. For an SME this process is doubly critical. Improvement in the sales order cycle time is vital for a company's success, while controlling credit and measuring payment is crucial for an SME to manage its cash flow efficiently and to gain customer satisfaction [2]. The process is thus simultaneously the firm's revenue engine and its principal source of working-capital risk.

The actors are the customer (restaurant or retailer), the sales representative, the sales manager who authorises non-standard discounts, warehouse and logistics staff, and the finance/collections function. The value delivered to the customer is the reliable supply of ordered goods at an agreed price and delivery date. The process has three principal outcomes: the order is fulfilled and paid; the order is cancelled because stock or price cannot be agreed; or the invoice remains unpaid and is passed to debt collection. Its performance is conventionally measured on order cycle time, order-error rate and days sales outstanding.

1.3 Process selection and current dysfunction

The process was selected on the standard criteria of importance, health and feasibility. Its importance follows from the fact that it generates all of the firm's revenue. Its poor health is well documented in the wholesale-distribution literature: in the majority of small and medium-sized trading companies the order process consists of a long chain of steps complicated by the need for approvals at various points, which significantly increases its duration and carries substantial risk of errors and inaccuracies during the transfer of order information and documents [3]. The consequences are familiar — incorrect pricing, excessive calls to the customer for information, order-processing delays and missed delivery dates — and their root cause is that a high number of manual transactions creates many opportunities for data-entry errors, while information stored in the system is not available in real time [2]. Feasibility is high because the process is high-volume, largely rule-governed and already partially digitised in spreadsheets and e-mail.

In line with the assessment brief, a single process is modelled and two distinct automation solutions are proposed against different tasks within it: a rule-based Robotic Process Automation (RPA) solution for the deterministic back-office tasks, and an agentic, LLM-based solution for the one cognitive task that rule-based automation cannot address.

2. Process Modelling

2.1 Notation and modelling approach

The process is modelled in Business Process Model and Notation (BPMN) 2.0. BPMN is a standard for business process modelling that provides a graphical notation based on traditional flowcharting techniques, designed to be readily understandable by all business stakeholders — analysts, developers and managers alike — and thereby to bridge the communication gap that frequently occurs between business process design and implementation [4]. Within the notation, pools represent the main participants in a process while lanes inside a pool show the activities of specific roles and define responsibilities in each part of the process [5].

Both an As-Is and a To-Be model are produced. Modelling the two in the same notation improves understanding of the current system and allows the desired system to be defined [6], and it is the redesign step — the elimination of non-value-adding activities before anything is automated — that produces most of the measurable gain; a comparable BPMN-based reengineering study raised throughput efficiency from as low as 40% to 100% by eliminating redundant steps and automating data recording [5]. Following the experience reported in an industrial BPMN case study, where a deliberately limited subset of BPMN constructs enabled process participants to communicate easily and to suggest changes themselves [7], the models here restrict themselves to pools and lanes, tasks with type markers, exclusive gateways, message and timer events, link events and message flows.

Because the end-to-end model exceeds a size that can be printed legibly on a single page, it is decomposed into two phases connected by BPMN link intermediate events, which act as off-page connectors: Phase A covers order intake and order handling, Phase B covers fulfilment, invoicing and collections. Two corresponding link events are semantically equivalent to a single sequence flow, so the decomposition does not alter process behaviour.

2.2 The As-Is process (Figures 1 and 2)

Orders arrive by telephone, e-mail or messaging application and are transcribed by hand into a spreadsheet. Where details are missing the representative telephones the customer and re-enters the order, producing the rework loop visible in Figure 1. Stock availability is confirmed verbally with the warehouse; if stock is short, a substitute is negotiated and the order may be cancelled. Pricing and discounts are applied manually from the customer's contractual terms, and any discount above 10% is routed to a manager for approval. A confirmation e-mail is then typed and sent.

Figure 2 shows the remainder. After picking and delivery, an invoice is prepared and sent manually. A 30-day payment term is represented by a timer event; when it expires the receivable is checked and, if unpaid, a reminder is typed and sent, with a 7-day timer governing the interval between reminders. After a third unsuccessful reminder the account is referred to debt collection. Every non-physical task in both diagrams carries the user-task marker: the process is almost entirely manual.

2.3 The To-Be process (Figures 3 and 4)

The To-Be model introduces a fourth lane, the Automation Layer, which holds the AI agent and the RPA bots. Order e-mails now enter an agent service task that extracts a structured order; an exclusive gateway then tests whether the extraction is both complete and above a confidence threshold. Above the threshold the order passes straight through to automated stock checking, rule-based pricing, ERP registration and confirmation; below it, the order is routed into the Sales Representative lane for review and correction. Substitute negotiation, discount approval above 10%, physical picking and delivery, and the final collections call remain deliberately human. In Phase B, invoicing and the reminder loop become unattended bots, and the hand-off to a human collections officer is triggered after two automated reminders rather than three.

One limitation of the notation is worth recording. BPMN is activity-centric, and a model written in it lacks constructs for expressing an agent's planning capability or the frame that constrains it [8]. The agent is therefore represented as an ordinary service task, and the frame that bounds its autonomy is made visible in the model in the only way the notation allows: through the explicit confidence gateway and the escalation path that leaves the Automation Layer for the human lane. This modelling compromise is revisited in Section 4.2.

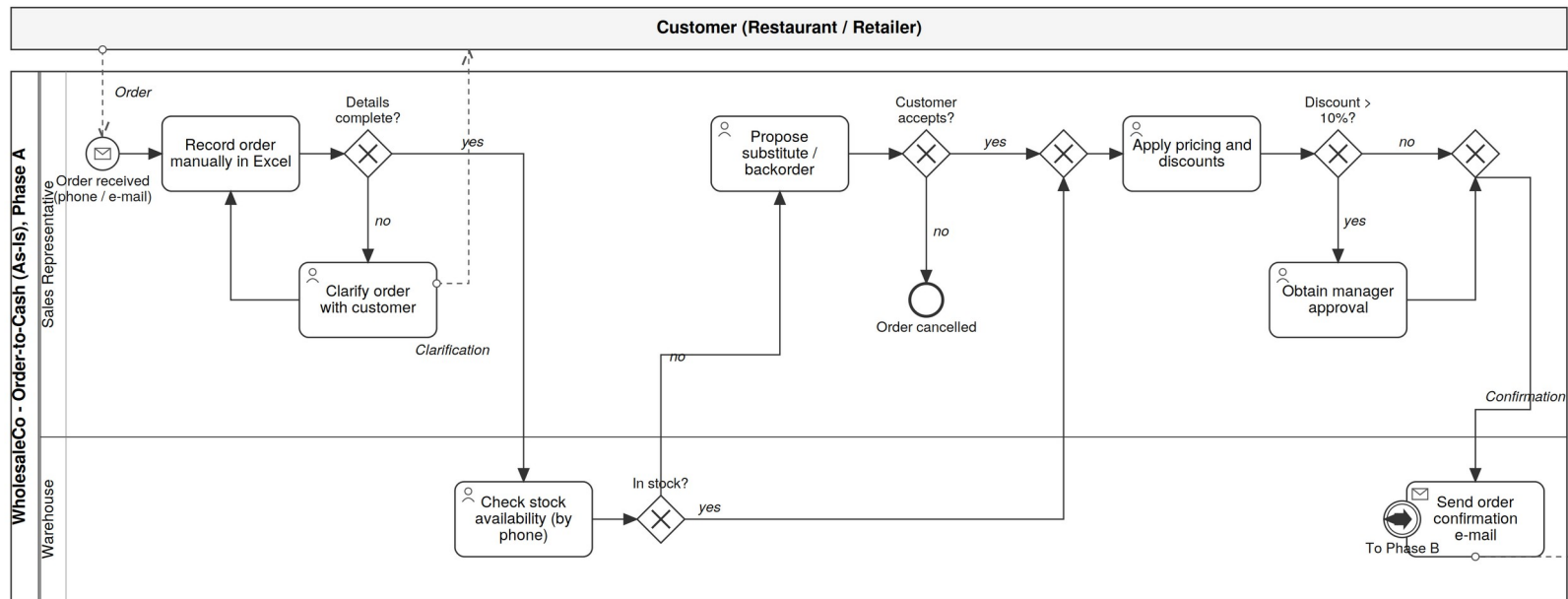


Figure 1. As-Is process, Phase A — order intake and order handling.

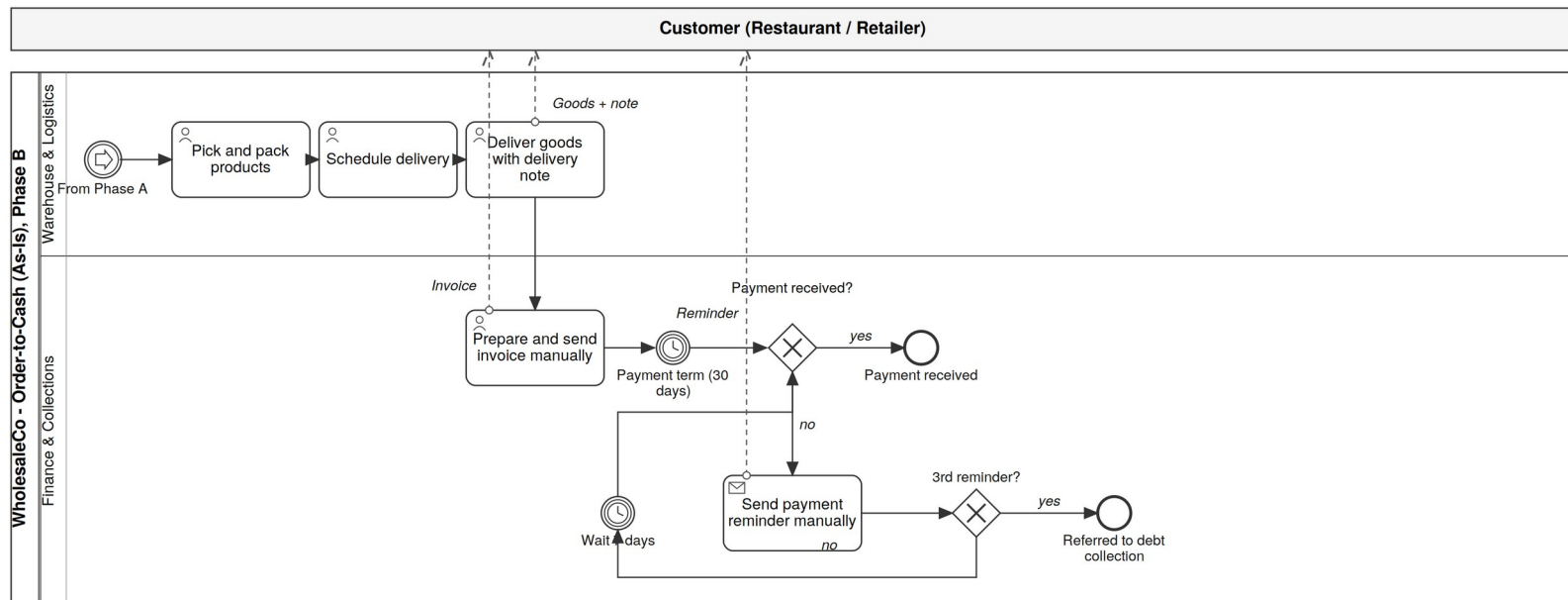


Figure 2. As-Is process, Phase B — fulfilment, invoicing and collections.

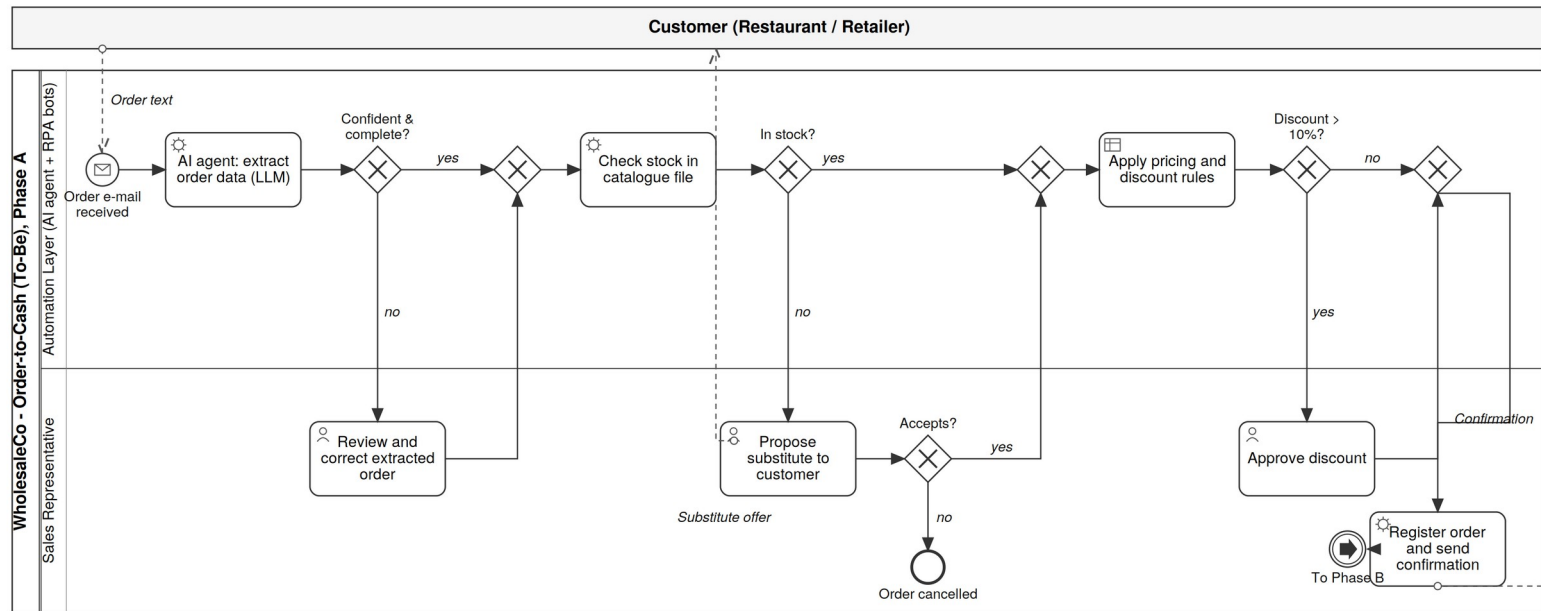


Figure 3. To-Be process, Phase A — agentic order intake with confidence-gated escalation.

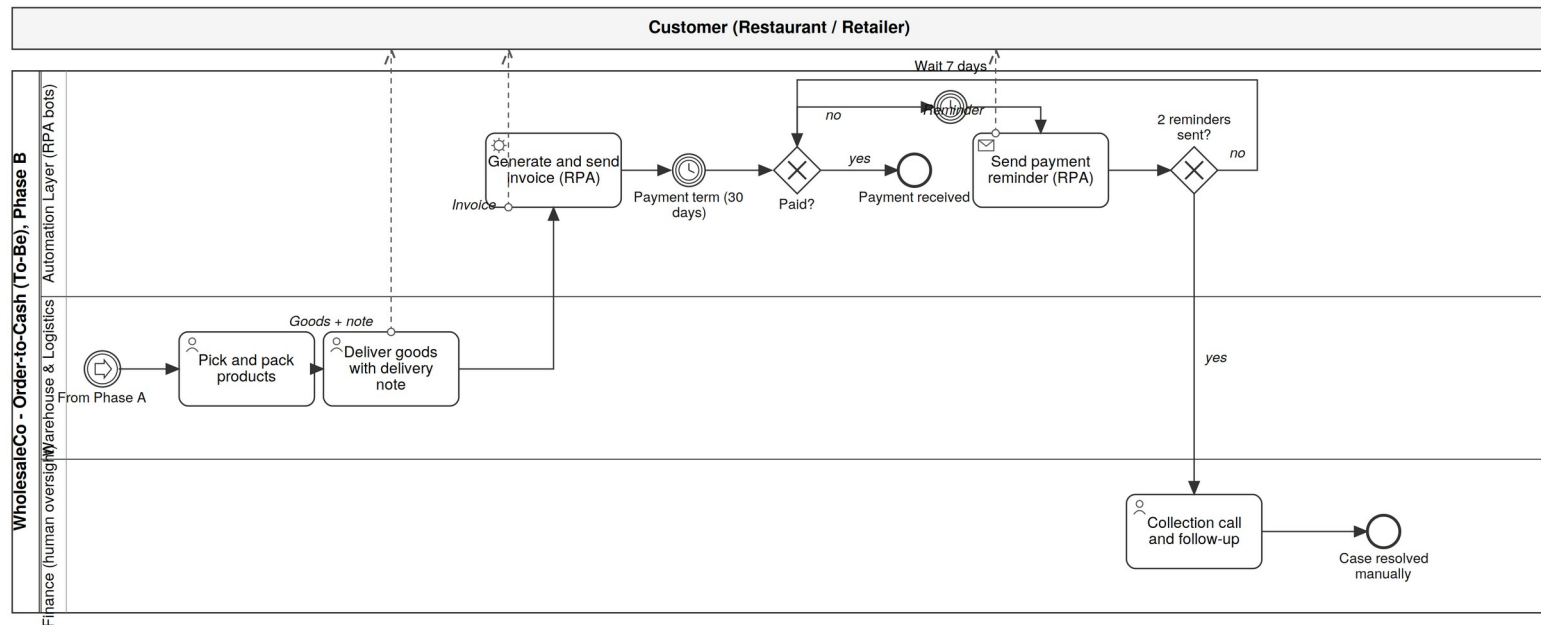


Figure 4. To-Be process, Phase B — automated invoicing, reminder loop and human collections hand-off.

3. Automation Potential Analysis

Each task of the As-Is model was assessed against the suitability characteristics identified in a systematic review of RPA adoption, namely that a process should be rule-based, mature, supported by structured and digital data, high-volume, routine, subject to few exceptions, and ideally spanning multiple systems [9]. Tasks failing the rule-based or structured-data tests are not thereby unautomatable; they are candidates for intelligent rather than robotic automation, because agentic AI can operate beyond a set of rules by sensing and reasoning about the current state of the process rather than following fixed scripts [8].

Task	Data	Verdict	Justification
T1 Receive and interpret order	Unstructured free text	Agentic AI	Not rule-based; wording, quantities and units vary by customer. Rule-based bots require structured documents stored digitally [9], so this task is the natural boundary between RPA and intelligent automation.
T2 Record order in system	Structured (post-T1)	RPA	Deterministic transcription; high volume, digital, few exceptions once T1 has produced a schema-conformant order.
T3 Clarify order with customer	Conversational	Human	Requires relationship judgement; low volume, high variability.
T4 Check stock availability	Structured, digital	RPA	Simple threshold comparison against the catalogue; routine and repeated for every line item.
T5 Propose substitute	Conversational	Human	Commercial negotiation; outcome not determinable from data alone.
T6 Apply pricing and discounts	Structured, digital	RPA	Pure if-then logic over contractual terms; decisions require no judgement and are therefore automatable through decision rules [9].
T7 Approve discount above 10%	Structured	Human	Retained deliberately as a governance control rather than a technical limitation.
T8 Send order confirmation	Templated	RPA	High-volume templated correspondence with a document attachment.
T9 Pick, pack and deliver	Physical	Out of scope	Physical handling; outside the reach of software automation.
T10 Prepare and send invoice	Structured, digital	RPA	Document generation from ledger data; mature, stable and high-volume.
T11 Monitor due dates, send reminders	Structured, digital	RPA	Routine and periodic; a scheduled trigger allows the bot to run without human interference [9].
T12 Collections call	Conversational	Human	Negotiation of payment plans; reputational risk if automated.

Table 1. Task-by-task automation assessment of the As-Is process.

3.1 Short-term and long-term implications, and risk assessment

In the short term the six RPA candidates offer rapid, low-risk gains. Reported quantitative benefits of successful RPA implementations include cost savings between 25% and 75% and substantial process-time reduction, alongside lower error rates, the ability of bots to work at any time, and greater standardisation [9]. A directly comparable study of order reception and processing in wholesale trading companies recorded a fall in order completion time from 4.85 to 0.33 working hours after the manual coordination steps were removed and an information system took them over [3].

In the longer term the picture is more demanding. Suitability depends on process maturity — the process should be subject to minimal change in the near future, so that outcomes are predictable and exceptions are less likely to arise [9] — and where that assumption fails the maintenance burden grows. This is not a hypothetical concern: high initial expectations and maintenance costs have already led to mixed experiences in RPA adoption [10]. The realistic long-term risks are therefore threefold. First, unfit process selection: attempting to automate manual, complex or highly fractional tasks is a recurring challenge, and either redesigning the process or choosing a fitter one is the better response [9]. Second, access and security: current security practices do not account for the existence of digital workers, and a

bot's access to systems and credentials must be governed explicitly [9]. Third, data incompatibility and undocumented automations, both of which the review identifies as common obstacles [9].

The agentic component carries a different risk profile. Practitioners interviewed about agent adoption anticipate efficiency, data-quality and compliance gains but caution against bias, over-reliance on automation, cybersecurity threats, job displacement and ambiguous decision-making [10]. These risks are addressed structurally in Section 4.2 rather than by assurances: the agent's autonomy is bounded, its uncertain cases are handed to a human, and every decision it takes is logged.

4. Automation Proposals

4.1 RPA-based automation

Target tasks: T2, T4, T6, T8, T10 and T11. Collectively these are the deterministic back-office spine of the process, and each satisfies the suitability criteria set out above.

Approach. Two unattended bots are proposed. An order-processing bot is event-driven: whenever a schema-conformant order becomes available it validates the customer against master data, checks and decrements stock, applies the contractual discount, flags discounts above the 10% threshold for manager approval, writes the order to the ERP ledger, generates a PDF invoice and sends a confirmation e-mail with the invoice attached. A collections bot is time-driven: it runs on a daily schedule, scans the ledger for invoices past their due date, respects a seven-day interval between reminders, sends the reminder, and after two reminders stops mailing and hands the account to a human officer. A scheduled trigger of this kind is precisely the pattern the literature identifies as most apt for automation, since a set moment in time can trigger the robot without any human interference [9].

Tools and technologies. The implementation uses Python with openpyxl for Excel and ERP-ledger access, reportlab for invoice generation, the standard smtplib and email libraries for correspondence (interchangeable with win32com for Outlook in a Windows environment), and the logging module for an audit trail; scheduling is delegated to Windows Task Scheduler or cron. Commercial alternatives — UiPath, Automation Anywhere, Blue Prism or Power Automate — would implement the same design through visual designers; Python was chosen for transparency and reproducibility in an academic submission. In either case the bot functions as a top layer providing integration between different systems [9], which is exactly its role here between the spreadsheet catalogue, the ERP ledger and the mail server.

Expected benefits. The direct effects are reduced process hours, a lower error rate, cost savings, availability outside working hours and standardisation of output [9]. The indirect effect is the more valuable one: converting to automated order processing allows a firm to realise efficiencies throughout the order-to-cash cycle, not merely in the automated step [2], because downstream tasks receive clean, timely data.

Potential challenges. The bots are brittle with respect to file and interface changes, so the spreadsheet schemas must be treated as contracts. Bot credentials require the same governance as human ones, given that access and security are among the key issues for RPA implementations [9]. Exception paths — an unknown customer, a corrupt ledger row — must fail loudly into a human queue rather than silently. Finally, employee and stakeholder resistance is a documented obstacle [9] and is mitigated here by leaving negotiation, approval and collections work explicitly with people.

4.2 Agentic (AI-driven) automation

Target task: T1, the interpretation of a free-text customer order. This is the bottleneck that blocks the entire RPA chain, and it is beyond rule-based automation by definition.

Approach. The solution is an order-intake agent built on the sense–decide–act structure by which agentic AI is characterised: sensing is the ability to analyse its environment and extract relevant insights, deciding is the ability to evaluate options and determine the next course of action, and acting is the execution of the chosen strategy through interfaces to other systems [8]. Concretely, the agent senses by reading the inbox together with the product catalogue and customer master; it decides by prompting a large language model for a schema-constrained extraction, grounding every extracted product against the catalogue by fuzzy matching, and combining the model's self-reported certainty with the worst catalogue-match score into a single confidence value; it acts by writing a structured order for the RPA bot, or by escalating to a human.

Design element	Purpose and justification
Product catalogue and customer master as agent	Grounds extraction in real data so that a plausible-sounding but non-existent product cannot enter the order.

Design element	Purpose and justification
context	
Strict JSON schema, temperature 0	Makes the agent's output machine-consumable by the downstream RPA bot and reduces variability.
Confidence threshold (0.80) and escalation path	Operationalises the open question of when to refrain from autonomy and return control to a human agent where the agent has insufficient confidence and is at risk of making mistakes; confidence bounds beyond which a process must escalate to human decision-makers are the proposed mechanism [11].
Escalation record naming the specific ambiguity	Gives the reviewing sales representative the reason, not just the failure, supporting the verification pattern in which one performer reviews and validates the output of another [8].
Three-attempt retry with exponential backoff, and an offline fallback parser	Provides the fallback mechanism that enables safe reversion if the external service is unavailable [10].

Table 2. Design elements of the agentic solution and their justification.

Framed autonomy. The design follows the principle that framing is the primary mechanism for ensuring process-awareness and goal alignment in an agentic system, imposing restrictions on the autonomy of agents through their knowledge and their goals [11]. The frame here consists of the catalogue and customer master that bound what the agent may recognise, the output schema that bounds what it may produce, and the confidence threshold that bounds when it may act at all. It also satisfies the practitioner recommendations for responsible agent adoption, which call for clear guardrails, defined human–agent collaboration with an identified point of human intervention on agent failure, and safe integration with fallback options [10].

Tools and technologies. Google Gemini (gemini-2.5-flash) is called over its REST interface with a JSON response type and temperature 0; catalogue grounding uses difflib sequence matching; the agent degrades to a deterministic offline parser when no API key is configured, so that the pipeline remains reproducible for assessment.

Expected benefits and challenges. The benefit is that removing this single manual step unlocks straight-through processing for the whole order-to-cash chain while keeping ambiguous cases under human control. The challenges are those of the technology itself: the stochastic nature of generative AI raises real governance concerns [10], hallucinated products are possible and are mitigated — not eliminated — by catalogue grounding, per-call latency and cost are incurred on every order, and customer correspondence sent to an external model raises data-protection questions that a production deployment would need to resolve.

5. Solution Implementation

Both proposals are implemented in Python as four executable scripts sharing a configuration module and a helper module. The architecture and the data flow between them are shown in Figure 5. The agent's output is the RPA bot's input, exactly as in the To-Be model: the AI handles the cognitive task, the bots handle the deterministic ones, and humans receive only the exceptions.

Figure 5. Solution architecture and data flow

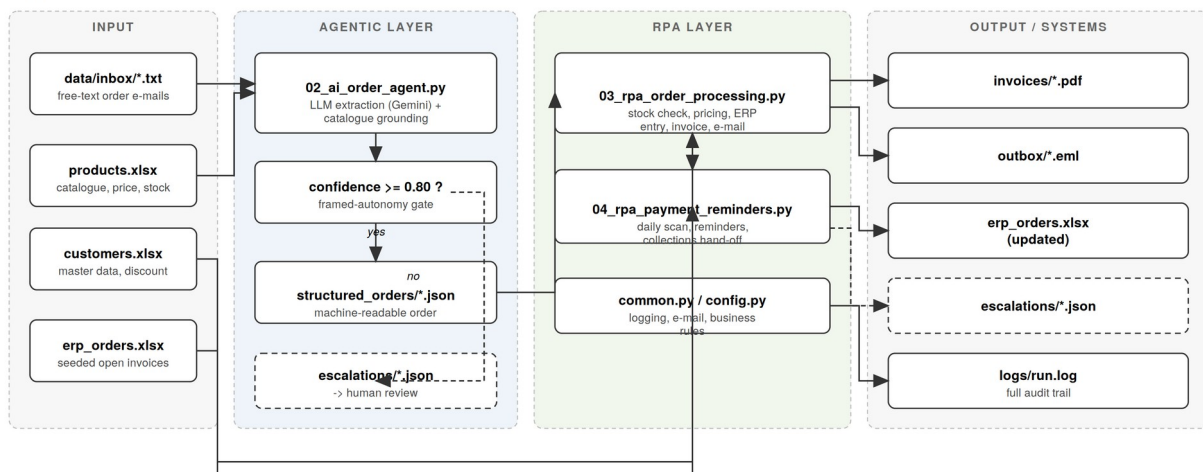


Figure 5. Solution architecture: agentic layer, RPA layer and the artefacts they produce.

Script	Solution	Function
01_setup_sample_data.py	Test fixture	Creates the catalogue, customer master, four free-text order e-mails and an ERP ledger seeded with two overdue invoices.
02_ai_order_agent.py	Agentic	LLM extraction, catalogue grounding, confidence scoring, escalation decision.
03_rpa_order_processing.py	RPA	Stock check, pricing, ERP entry, PDF invoice, confirmation e-mail.
04_rpa_payment_reminders.py	RPA	Overdue scan, reminder interval, collections hand-off.

Table 3. Implementation components.

Business rules are held in a single configuration file — the 10% discount threshold, the 30-day payment term, the 7-day reminder interval, the two-reminder ceiling and the 0.80 confidence threshold — so that every parameter in the BPMN models has one authoritative representation in code. E-mail is simulated by writing RFC-822 .eml files to an outbox directory, which keeps the demonstration portable; a single configuration flag switches the same function to SMTP.

Demonstration. The sample data is constructed so that a single run exercises every branch of the To-Be model, including the exception paths, which are the branches most often left undemonstrated.

Scenario	Path exercised and observed result
email_01 (Anatolia Restaurant)	Clean order, confidence 0.95 — straight-through processing to ORD-1003, invoice INV-1003, confirmation e-mail sent.
email_02 (Green Valley Foods)	Clean order but a 12% contractual discount exceeds the threshold — registered as ORD-1004 with status PENDING_APPROVAL for a manager.
email_03 (Bosphorus Kebab House)	50 bags of rice requested against 35 in stock — escalated to sales for a substitute proposal; no order registered.
email_04 (Liffey Mini Market)	"some cheese and the usual oil" — no usable quantities, confidence 0.35 below threshold, escalated to human review with the specific missing quantities listed.
Seeded INV-1001	15 days overdue, no prior reminders — reminder 1 of 2 sent automatically.
Seeded INV-1002	30 days overdue, two reminders already sent — automated path exhausted, escalated to human collections.

Table 4. Demonstration scenarios and observed results.

Figure 6. Console output of a complete end-to-end run (MOCK mode)

```
[19:33:53] INFO      setup: Seeded ERP ledger with 2 historical invoices -> erp_orders.xlsx
[19:33:53] INFO      ai_agent: AI order agent starting in MOCK (offline parser) mode
[19:33:53] INFO      ai_agent: --- Processing email_01 anatolia.txt ---
[19:33:53] INFO      ai_agent: Confidence 0.95 >= 0.80 -> structured order written: email_01_anatolia.json
[19:33:53] INFO      ai_agent: --- Processing email_02 greenvalley.txt ---
[19:33:53] INFO      ai_agent: Confidence 0.95 >= 0.80 -> structured order written: email_02_greenvalley.json
[19:33:53] INFO      ai_agent: --- Processing email_03 bosphorus.txt ---
[19:33:53] INFO      ai_agent: Confidence 0.95 >= 0.80 -> structured order written: email_03_bosphorus.json
[19:33:53] INFO      ai_agent: --- Processing email_04 liffey_ambiguous.txt ---
[19:33:53] WARNING   ai_agent: ESCALATED to human review: email_04_liffey_ambiguous_ESCALATED.json (reasons: Missing quantity for '...'
[19:33:53] INFO      ai_agent: AI order agent finished.
[19:33:54] INFO      rpa_orders: RPA order bot: 3 structured order(s) to process
[19:33:54] INFO      rpa_orders: E-mail 'Order confirmation ORD-1003' -> orders@anatolia-restaurant.ie (20260730_193354_271111_Orde...
[19:33:54] INFO      rpa_orders: ORD-1003 registered: net 697.00, gross 721.74, approval_required=NO
[19:33:54] INFO      rpa_orders: E-mail 'Order confirmation ORD-1004' -> buying@greenvalleyfoods.ie (20260730_193354_276736_Order_c...
[19:33:54] INFO      rpa_orders: ORD-1004 registered: net 1026.90, gross 985.00, approval_required=YES
[19:33:54] WARNING   rpa_orders: Stock shortage on email_03_bosphorus.json -> escalated (Baldo Rice 25kg (asked 50, in stock 35))
[19:33:54] INFO      rpa_orders: ERP ledger and stock updated. RPA order bot finished.
[19:33:54] INFO      rpa_reminders: Reminder bot run for 2026-07-30 - scanning 4 ledger rows
[19:33:54] INFO      rpa_reminders: E-mail 'Payment reminder - invoice INV-1001' -> purchasing@liffeymarket.ie (20260730_193354_543...
[19:33:54] WARNING   rpa_reminders: INV-1002: 2 automatic reminders exhausted -> ESCALATED to human collections
[19:33:54] INFO      rpa_reminders: Wrote 1 collections escalation(s) -> collections_escalations.json
[19:33:54] INFO      rpa_reminders: Reminder bot finished: 1 reminder(s) sent, 1 escalation(s).
```

Figure 6. Console output of a complete end-to-end run, showing both straight-through and escalated cases.

Verified artefacts. A run produces PDF invoices with correct line totals, discount and VAT arithmetic; .eml confirmation and reminder messages; an updated ERP ledger in which ORD-1004 carries PENDING_APPROVAL and INV-1002 carries ESCALATED_TO_COLLECTIONS; JSON escalation records for the ambiguous order, the stock shortage and the exhausted reminder cycle;

decremented stock levels in the catalogue; and a complete run log. Together these constitute the evidence that the two automations behave as the To-Be model specifies.

6. Conclusion

This report identified the Order-to-Cash process of a B2B food wholesaler as a high-importance, low-health and high-feasibility candidate for automation, modelled it in BPMN in both As-Is and To-Be forms, assessed every task against established suitability criteria, and proposed and implemented two complementary solutions. The division of labour between them is the central argument: RPA takes the deterministic tasks for which all scenarios can be anticipated and scripted, the agent takes the single cognitive task that defeats rule-based automation, and the human keeps negotiation, approval and relationship work. The confidence-gated escalation is what makes that division safe rather than merely convenient.

Two limitations should be acknowledged. The ERP, the mail server and the warehouse are simulated by spreadsheets and files, so the integration risk of a real deployment is not represented; and the agent's accuracy was demonstrated on four constructed e-mails rather than measured on a representative sample, so no claim about extraction accuracy in production is made here. A natural extension would be to close the loop by monitoring the executing process, since it is process mining that lays the foundations on which agents can sense process states, reason about improvement opportunities and act to maintain performance [8] — moving the solution from automation towards the autonomy that agentic BPM envisages.

References

- [1] N. B. Ilchenko, “Management of business processes at a wholesale enterprise,” *Business Inform*, no. 12, pp. 270–279, 2016.
- [2] B. Kocaoglu and A. Z. Acar, “Process development in customer order information systems to gain competitive advantage: a SME case study,” *International Journal of Logistics Systems and Management*, vol. 23, no. 2, pp. 209–230, 2016.
- [3] G. Mutanov, S. Ziyadin and A. Serikbekuly, “Application of system-dynamic modeling to improve distribution logistics processes in the supply chain,” *Communications – Scientific Letters of the University of Zilina*, vol. 22, no. 3, pp. 29–39, 2020.
- [4] M. von Rosing, S. White, F. Cummins and H. de Man, “Business Process Model and Notation — BPMN,” in *The Complete Business Process Handbook*. Elsevier, 2015, pp. 429–450.
- [5] U. F. Warvana and W. Suharso, “Business process reengineering of the book lending and returning system for library using BPMN,” *Bit-Tech*, vol. 8, no. 1, pp. 159–169, 2025.
- [6] L. Prades, F. Romero, A. Estruch, A. García-Domínguez and J. Serrano, “Defining a methodology to design and implement business process models in BPMN according to the standard ANSI/ISA-95 in a manufacturing enterprise,” *Procedia Engineering*, vol. 63, pp. 115–122, 2013.
- [7] M. zur Muehlen and D. T. Ho, “Service process innovation: a case study of BPMN in practice,” in *Proc. 41st Hawaii International Conference on System Sciences (HICSS)*, 2008.
- [8] M. Dumas, F. Milani and D. Chapela-Campa, “Agentic business process management systems,” Springer, 2026. doi: 10.1007/978-3-032-13426-4_1.
- [9] D. A. da Silva Costa, H. São Mamede and M. Mira da Silva, “Robotic process automation (RPA) adoption: a systematic literature review,” *Engineering Management in Production and Services*, vol. 14, no. 2, pp. 1–12, 2022.
- [10] H. Vu, N. Klievtsova, H. Leopold, S. Rinderle-Ma and T. Kampik, “Agentic business process management: practitioner perspectives on agent governance in business processes,” in *Proc. 23rd International Conference on Business Process Management*, Springer, 2025, pp. 29–43.
- [11] D. Calvanese et al., “Agentic business process management: a research manifesto,” *Information Systems*, vol. 140, art. 102738, 2026.